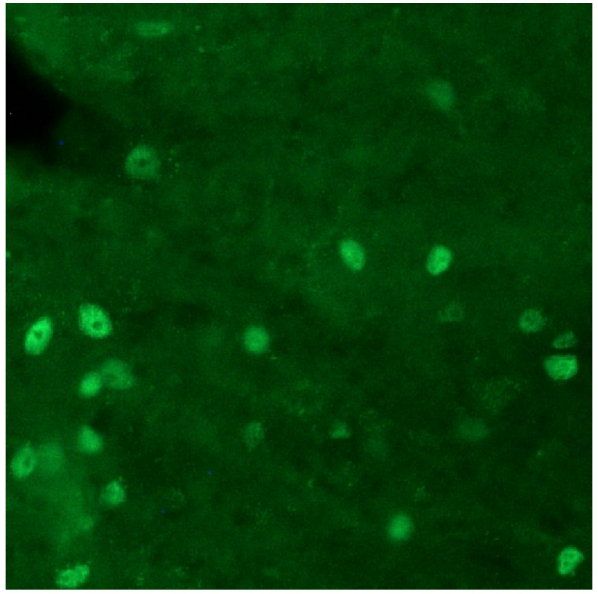
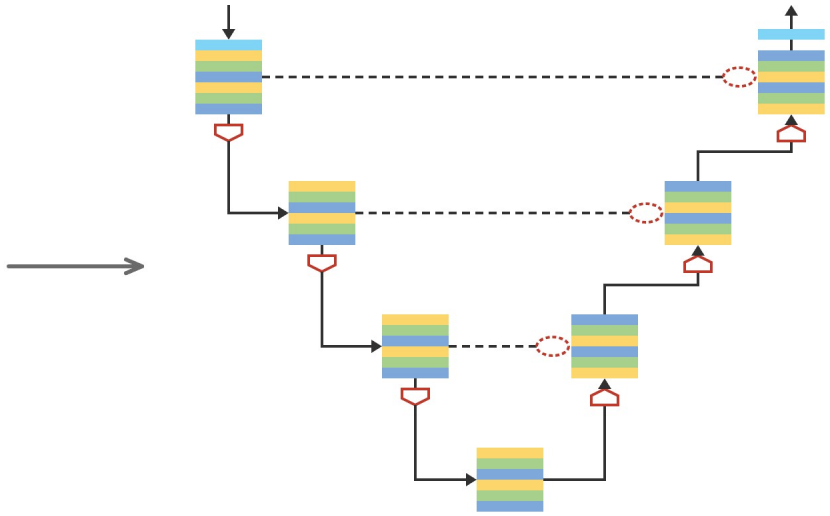
Segmentation and counting of c-FOS nuclei

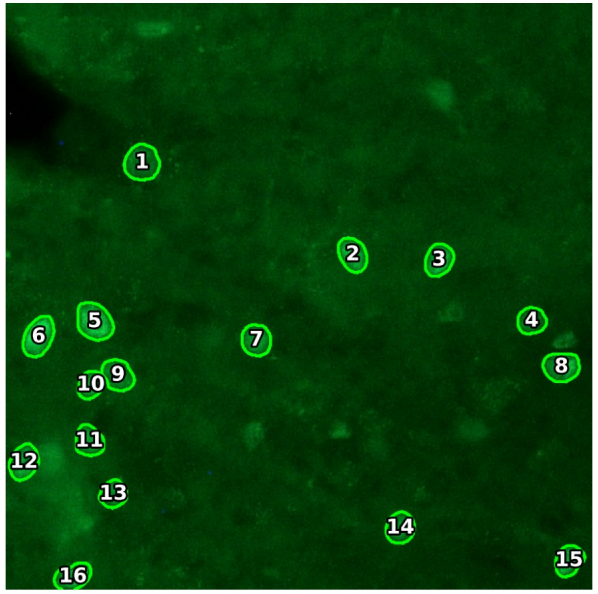
Fluorescent Neuronal Cells v2, Green collection (Clissa et al., 2024)



Test image



c-ResUNet-light + post-processing



16 cells found, 16 annotated

Ali Rammal · October 2026

Objective

Today: manual counting

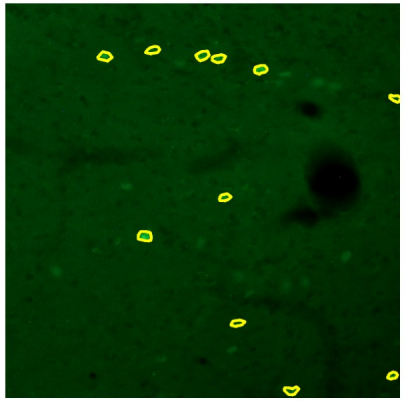
Slow	Image by image
Subjective	On faint cells
Variable	Across annotators and over time

Model: automatic counting

Automation	A few seconds per image
Same criterion	The same rule for every image
Interpretability	Every counted cell has its own outline

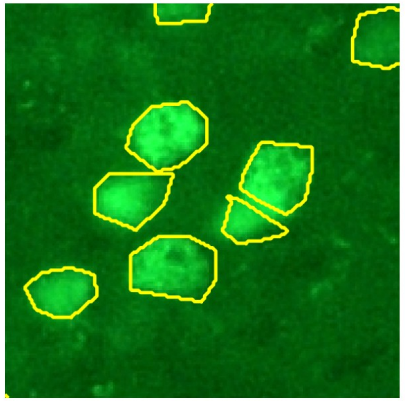
Challenges

Yellow: annotated cell · dashed white: unannotated object



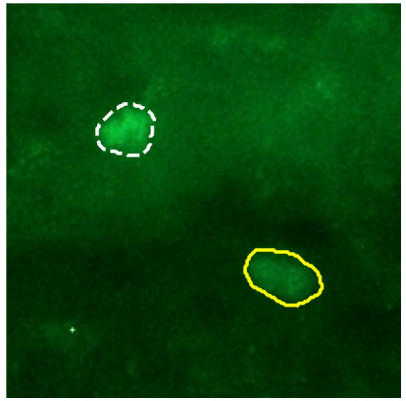
Imbalance

Cells of ~30 px: 0.6% of the pixels



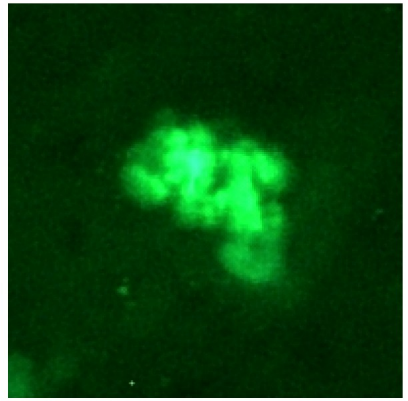
Clustered cells

Five cells a few pixels apart



Subjective labels

The unannotated object is brighter



Artifacts

Large, irregular blob, not annotated

The data

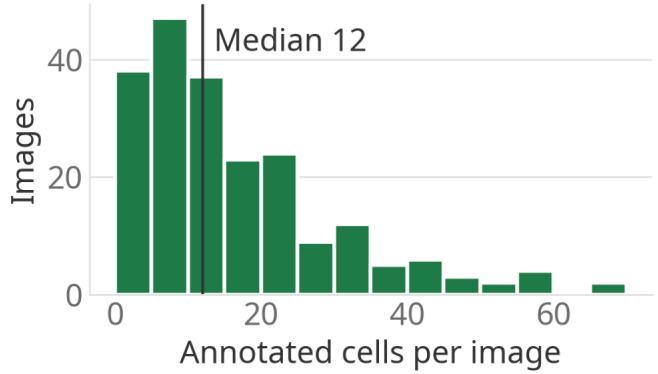
Split

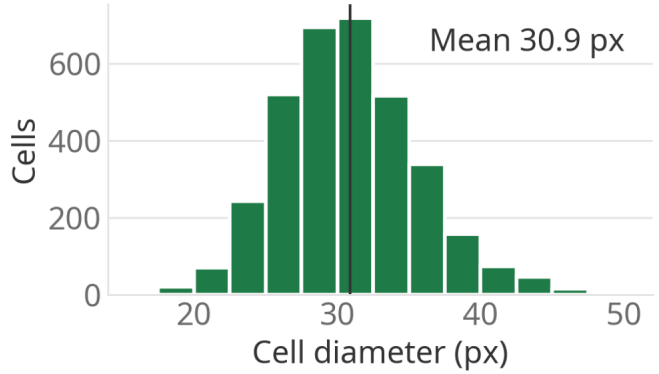
Set	Images	Cells	Use
Training	169	2690	Network training
Validation	43	731	Checkpoint selection and checks
Test	70	1172	Final evaluation only

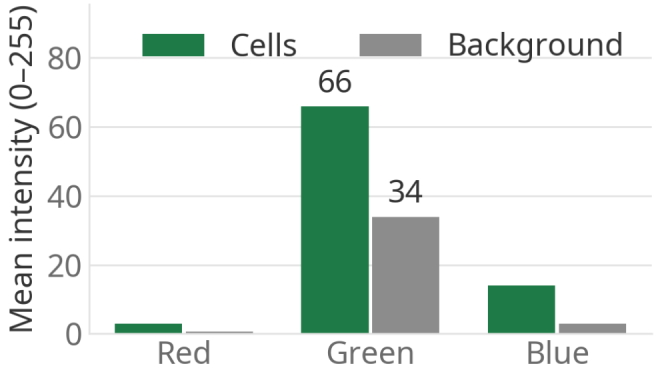
Training and validation: 80/20, stratified by number of cells

Characteristics

- 1200 × 1600 px, signal in the green channel
- From 0 to 67 cells per image (median 12)
- Cells ~30 px in diameter: 0.6% of the pixels
- Masks cleaned of 16 fragments of 1–2 px







The pipeline

Network
c-ResUNet-light: per-pixel probability

→

Threshold + area
Threshold 0.5, hole filling, removal of objects < 300 px

→

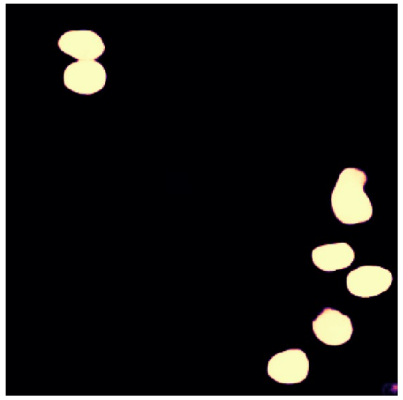
Watershed
Separates touching cells (seeds ≥ 20 px)

→

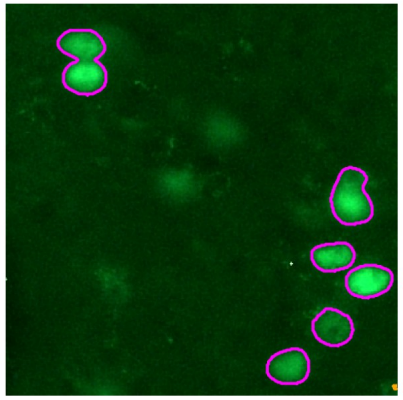
Classifier
Retains objects with P ≥ 0.51

→

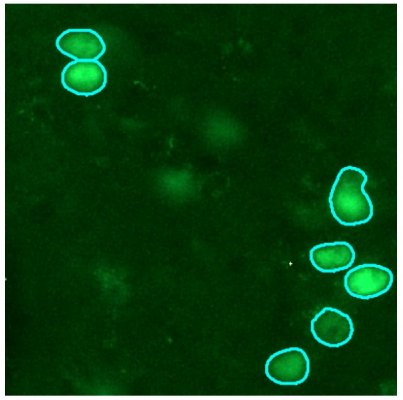
Result
Separated cells and count



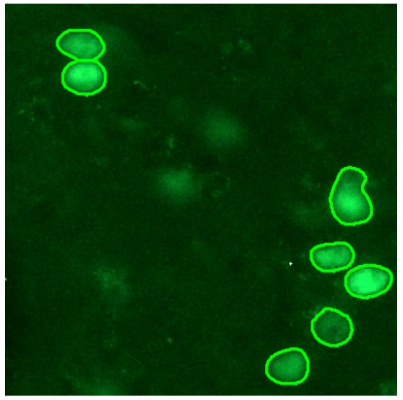
Cell probability, from 0 to 1



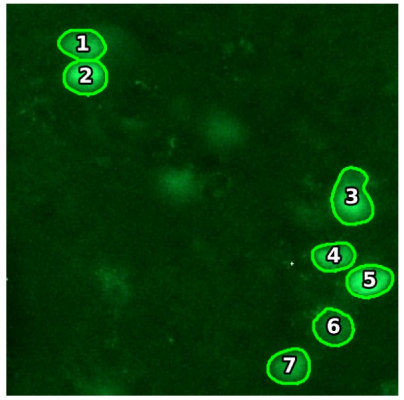
6 objects after threshold and area



1 merger separated: 7 objects

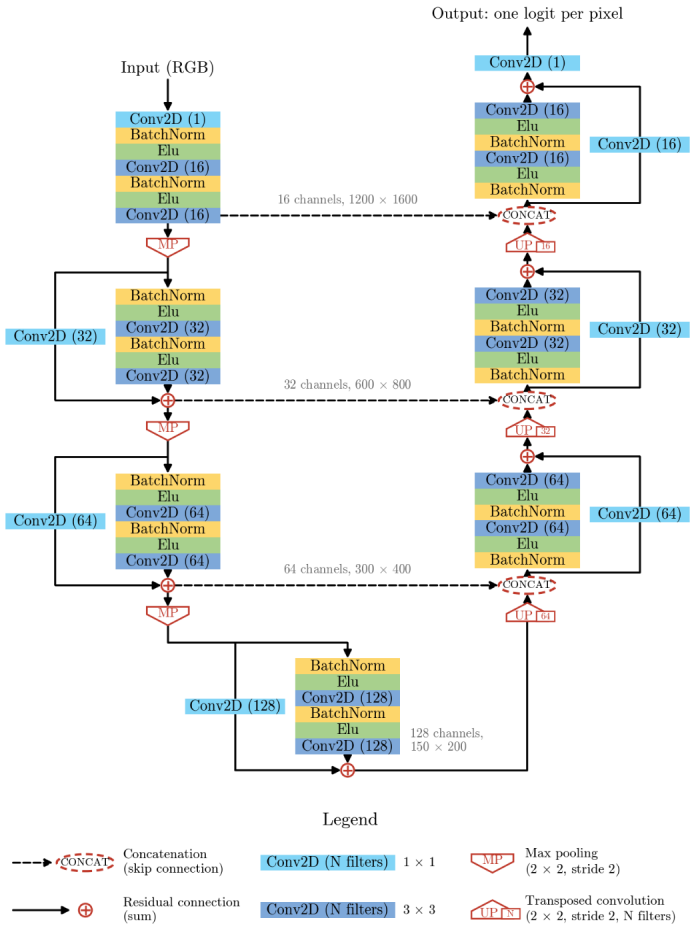


The classifier retains them all



7 cells counted, 7 annotated

The network: c-ResUNet-light



Architecture

Base	ResUNet (Zhang et al., 2018) + learned color conversion (Morelli et al., 2021)
5 × 5 block	Removed: no gain on Green, 2.6× fewer parameters
Size	0.50 M parameters, receptive field ~96 px
Output	One logit per pixel

Training

Augmentation	512 × 512 crops, flips, rotations, brightness and contrast
Loss	BCE + soft Dice
Optimizer	AdamW, cosine learning rate
Duration	100 epochs on a Kaggle T4 GPU (~1 h)
Checkpoint	Epoch 78: validation Dice 0.762, close to the training Dice (no overfitting)

Post-processing

Threshold 0.5

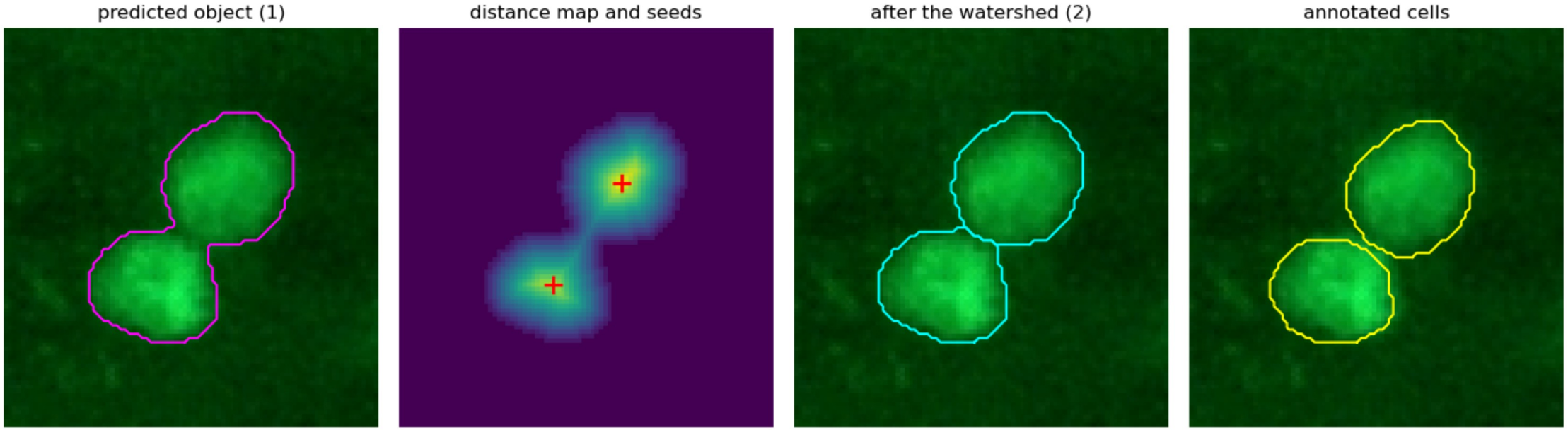
The network is almost always confident: no threshold from 0.3 to 0.7 does better.

Minimum area 300 px

Below it, almost no object is a cell: 3 out of 117 in training.

Watershed, seeds ≥ 20 px

Separates touching cells without splitting single ones.



The object classifier

Features and weights

Mean probability	+0.91
Relative contrast	+0.72
Solidity	+0.33
Number of objects	−0.32
Local background	−0.27
Contrast	+0.24
Area	−0.22
Intensity	+0.16
Distance to neighbor	−0.11
Eccentricity	+0.10

Logistic regression

Retains the object if P(cell) ≥ 0.51

Out-of-fold data

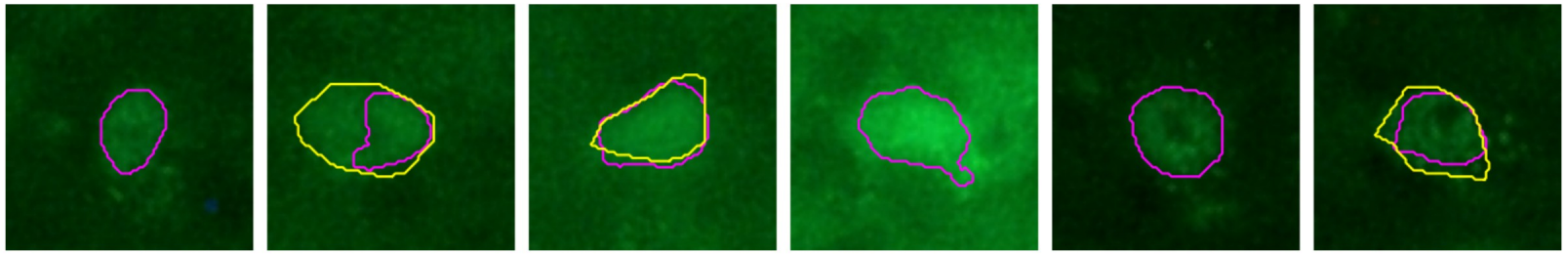
Objects from images the network never saw

Border

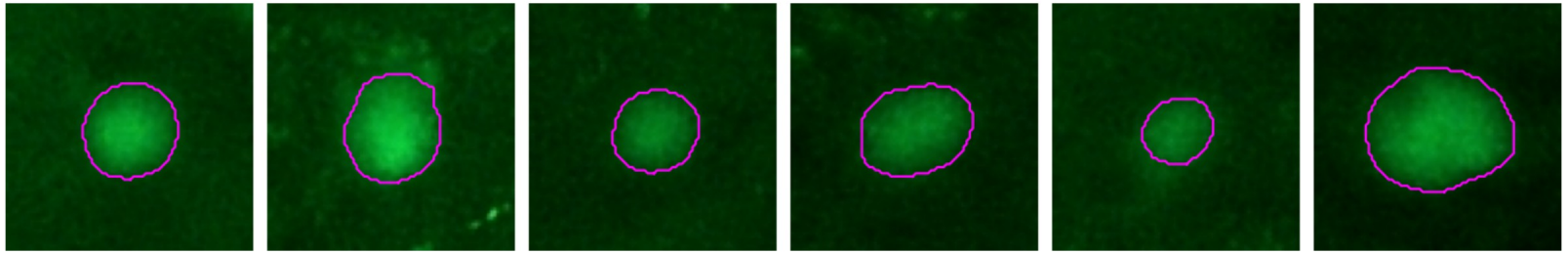
Objects on the border are never evaluated

Removed by the classifier

Magenta: evaluated object · yellow: annotated cell



Retained but not annotated



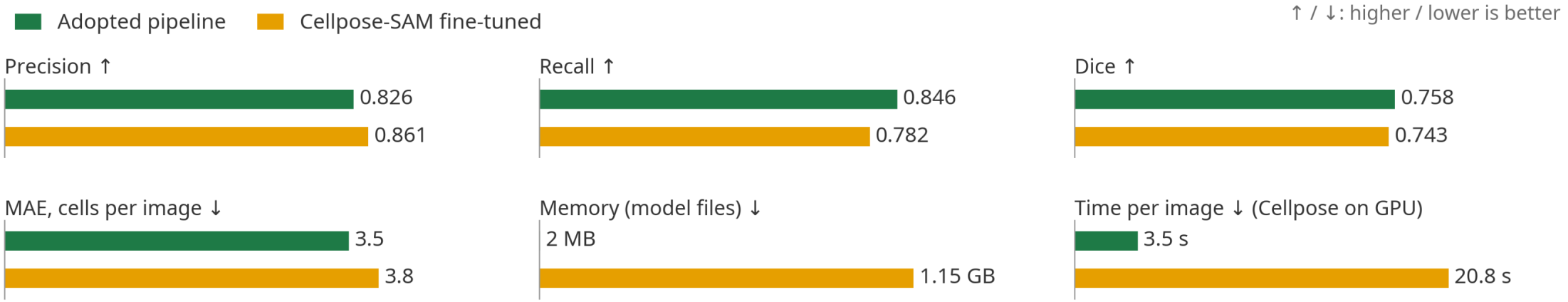
Decision rule

> **A step is adopted only if the F1 improves demonstrably, without worsening Dice or MAE**
>
> Demonstrably: the 95% interval of the difference lies entirely above zero (paired bootstrap over the images)



Test results: 70 images, 1172 cells

Method	Object F1 [95% CI]	Precision	Recall	Dice	MAE
Adopted pipeline	0.836 [0.811, 0.859]	0.826	0.846	0.758	3.5
Cellpose-SAM fine-tuned	0.819	0.861	0.782	0.743	3.8
Cellpose-SAM zero-shot	0.574	0.412	0.944	0.579	21.6
Robust threshold (median + 6.5 MAD)	0.506	0.441	0.595	0.579	9.8
Otsu	0.130	0.082	0.323	0.018	54.3

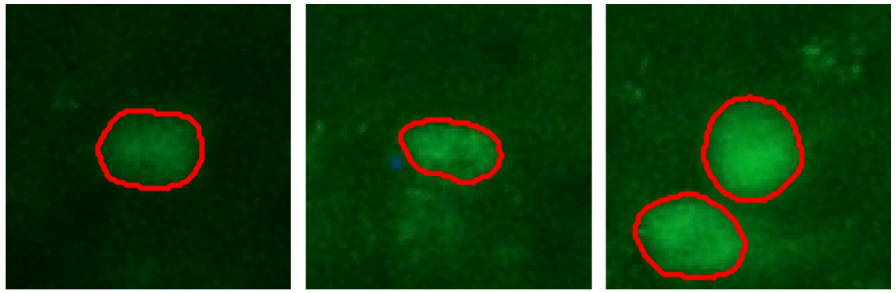


Errors

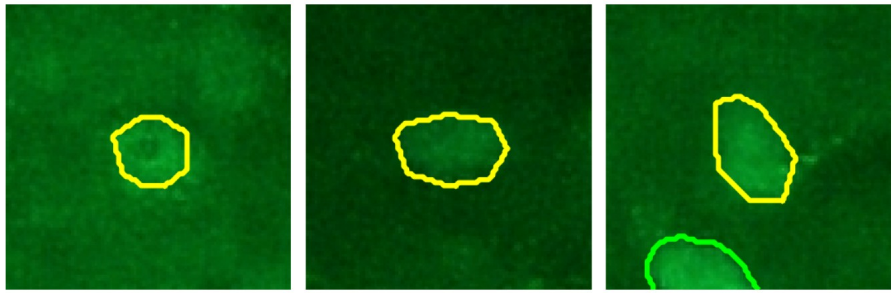
92% of the errors involve whole cells, not outlines

356 of 389 errors, mostly on faint cells, where the protocol calls for a conservative approach

Counted but not annotated · 199



Annotated but not found · 157 · in green: cell found



Conclusions

Result

Lightweight network (0.5 M parameters, ~1 h of training, 3.5 s per image on CPU) with simple post-processing: on par with fine-tuned Cellpose-SAM

Limitations

Faint cells: subjective decision and a single annotation per image. Touching cells only partly separated: 11 mergers remain

Future work

- Recognizing nuclei in any marker color (e.g. blue instead of green), by permuting the color channels in training
- Transfer learning to the Yellow and Red collections
- Green (nucleus) and yellow (cytoplasm) of the same section in a single image: segmenting both nuclei and cytoplasm

Code, notebooks, results: github.com/alirammal-ux/FNC_segmentation